

Models for Aspect-Based Sentiment Analysis (ABSA)

Model	Type	Purpose
GPT-5.5	Large Language Model (LLM)	Strong baseline for aspect extraction and sentiment analysis
BERT	Transformer Encoder	Traditional NLP baseline
RoBERTa	Improved BERT	Enhanced transformer for sentiment classification

GPT-5.5

Advantages

- Excellent at extracting aspects without additional training.
- Performs well in zero-shot and few-shot settings.
- Understands complex and long customer reviews.
- Can generate structured outputs such as JSON.
- Strong reasoning ability.

Limitations

- Requires API access and has usage costs.
- Output format may vary if prompts are not carefully designed.
- Fine-tuning is generally not available.

Possible Solutions

- Use a fixed prompt template.
- Force JSON output.
- Set a low temperature (e.g., 0–0.2) for consistent responses.
- Use few-shot prompting with examples.

BERT

Advantages

- Open-source and free to use.
- Easy to fine-tune for specific datasets.
- Fast training compared to larger language models.
- Widely used as a standard baseline in NLP research.

Limitations

- Cannot extract aspects effectively without training.
- Limited understanding of long-range context.
- Performance decreases on long and complex reviews.

Possible Solutions

- Fine-tune BERT on an ABSA dataset.
- Use domain-specific customer review datasets.
- Split long reviews into smaller sentences before processing.

RoBERTa

Advantages

- Improved version of BERT.
- Better contextual understanding.

- Higher accuracy in sentiment classification.
- Easy to fine-tune for downstream tasks.
- Frequently achieves better performance than BERT.

Limitations

- Does not perform aspect extraction effectively without training.
- Requires more computational resources than BERT.
- Longer training time.

Possible Solutions

- Fine-tune on ABSA datasets.
- Use mixed-precision training to reduce GPU memory usage.
- Combine with an LLM for aspect extraction.

Research Framework:

Dataset



Preprocessing



Run GPT-5.5

Run BERT

Run RoBERTa



Collect Predictions



Compare with Ground Truth



Calculate Accuracy, Precision, Recall, F1-score



Measure Inference Time



Compare Cost



Analyze Results & Write the Research Paper